



GUIDING DEMENTIA CONSULTATIONS IN PRIMARY CARE USING LARGE LANGUAGE MODELS AND EVIDENCE-BASED COMMUNICATION FRAMEWORKS *

MSc Semester Project Report

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ABSTRACT

Dementia care in primary care settings presents significant diagnostic and communication challenges. Clinicians must navigate complex consultations involving cognitive decline, diagnostic uncertainty, emotional distress, and caregiver concerns. More than 55 million people worldwide live with dementia, with nearly 10 million new cases annually. Primary care clinicians are often the first point of contact, yet existing artificial intelligence (AI) systems focus mainly on documentation automation, providing limited support for communication-centered workflows. The aim of this project was to develop a guided consultation assistant using large language models (LLM) and evidence-based communication frameworks. This project was conducted in collaboration with the Ariadne Labs and the Harvard T.H. Chan School of Public Health.

This project developed a standalone web-based clinical support system combining retrieval-augmented generation with large language models, grounded in the Ariadne Labs Essential Communications Toolkit. The system integrated structured workflow navigation across three consultation phases: recognition, evaluation, and diagnosis. It also incorporated the Stuck Points Framework for addressing relational communication obstacles. Implementation used open-source web technologies with Firebase backend and support for multiple LLM providers.

The system successfully integrated the Navigation Map structure with LLM-powered conversational guidance, enabling source-faithful responses grounded in evidence-based resources. The Stuck Points Framework was implemented as a specialized module for relational support, while the dual-mode architecture allowed comparison between different LLM providers while maintaining evidence grounding.

This work demonstrates the feasibility of combining large language models with structured communication frameworks to support primary care clinicians in dementia consultations. The findings establish that retrieval-augmented generation can provide clinically relevant, source-grounded guidance for communication-centered workflows, supporting further development of communication-centered clinical AI tools.

Keywords large language models · dementia care · clinical decision support · communication frameworks · retrieval-augmented generation

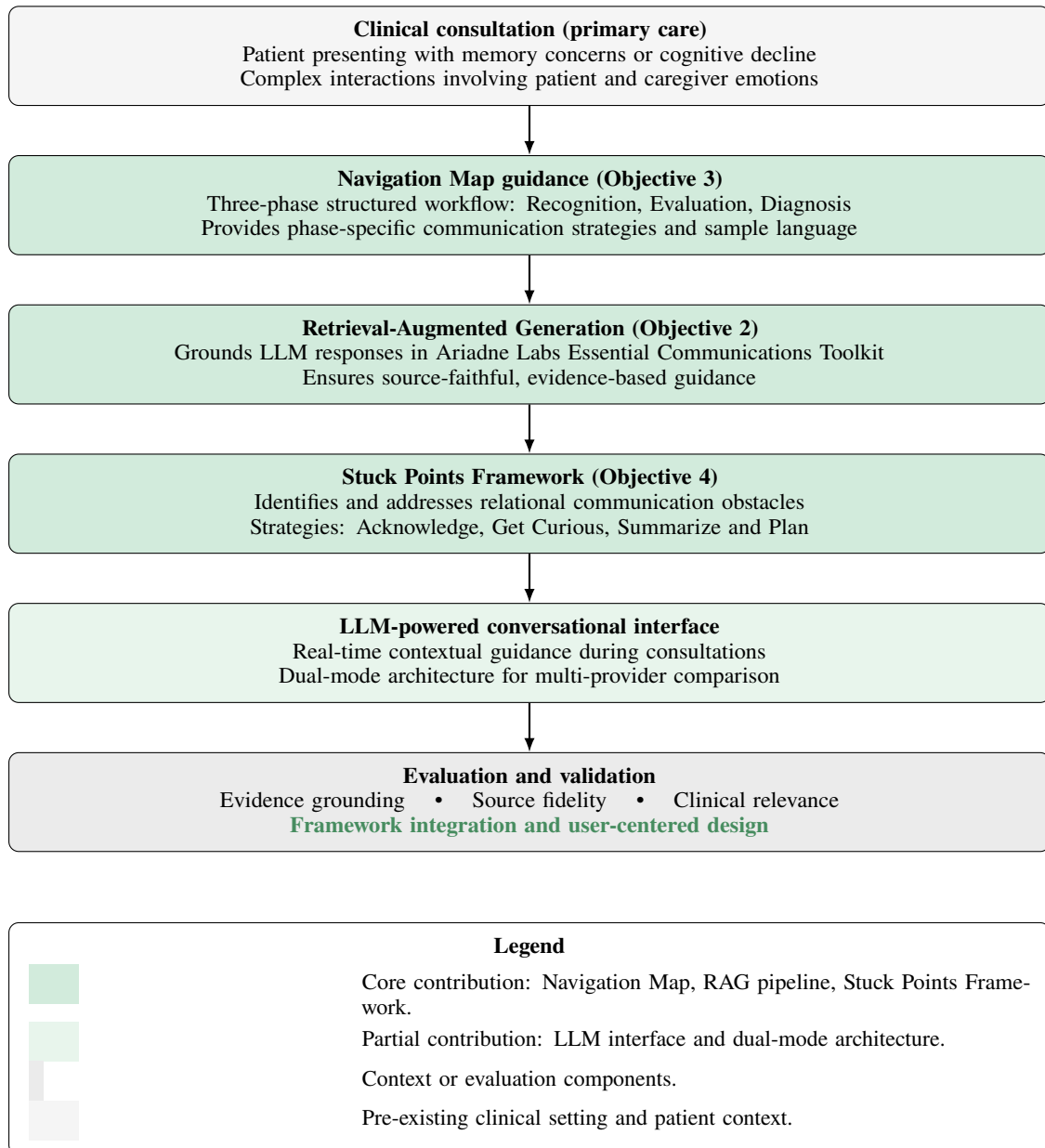


Figure 0.1: End-to-end overview of an LLM-guided dementia consultation system. Box fill indicates contribution level; green text marks key thesis contributions.

1 Introduction

1.1 Background and Research Gap

Dementia represents a growing global public health challenge associated with aging populations and increasing pressure on healthcare systems. With over 55 million people affected worldwide and nearly 10 million new cases annually, dementia is among the leading causes of disability and dependency in older adults [1]. Beyond the substantial emotional, social, and economic burden on patients and caregivers, early recognition and appropriate clinical management remain critical for improving outcomes and care planning.

Primary care clinicians play a central role in dementia care, serving as the first point of contact for patients and families. However, dementia consultations are particularly challenging due to diagnostic uncertainty, the need to integrate multiple assessment dimensions, and significant communication demands. Discussions regarding cognitive decline, loss of autonomy, or care planning may generate fear, denial, or emotional distress among patients and caregivers. Clinicians must balance clinical reasoning with empathetic communication while adapting their approach to sensitive interpersonal dynamics.

While LLMs have enabled various clinical support applications, existing AI systems primarily focus on documentation automation or generalized decision support. A critical gap remains: most tools do not actively guide clinicians through structured consultations or address relational communication challenges specific to dementia care. Concerns regarding hallucinations, explainability, and integration with clinical workflows further limit adoption in sensitive settings.

1.2 Research Aim and Objectives

This project addresses the identified gap by developing a guided consultation assistant that combines LLMs with evidence-based communication frameworks to support clinicians throughout dementia consultations.

To achieve this aim, the following objectives were pursued:

1. To review existing approaches for AI-assisted clinical support and dementia communication tools, identifying gaps in guidance, evidence grounding, and evaluation (Section 2).
2. To design and implement a retrieval-augmented generation system integrating the Ariadne Labs Essential Communications Toolkit with LLMs (Section 3).
3. To develop structured workflow navigation aligned with the Navigation Map framework across Recognition, Evaluation, and Diagnosis phases.
4. To implement the Stuck Points Framework for identifying and addressing relational communication obstacles.

1.3 Contributions

This project makes the following contributions:

- A standalone web-based clinical support system providing real-time conversational guidance grounded in evidence-based resources.
- Integration of the Navigation Map structure enabling structured workflow navigation across dementia consultation phases.
- A specialized Stuck Points Framework module for relational communication support during sensitive discussions.

These contributions demonstrate the potential for combining LLMs with structured communication frameworks to support communication-centered clinical workflows in dementia care.

2 Related Work

Recent advances in LLMs have enabled a new generation of clinical support tools designed to improve healthcare workflows and assist clinical reasoning. Existing approaches span AI-assisted documentation, clinical decision support systems, and LLM-based medical assistants. While these systems demonstrate growing feasibility for AI-assisted healthcare, important gaps remain regarding communication-centered workflows and specialized domains such as dementia care.

2.1 AI-Assisted Clinical Documentation

A significant body of work has focused on automating clinical documentation through AI scribes and note-taking systems. Platforms such as Doctolib and Tandem have demonstrated that AI can effectively capture and summarize medical encounters, highlighting key information for clinical review. Similarly, Suki.ai has developed voice-activated AI assistants for clinical documentation. These tools have shown strong adoption rates due to their focus on reducing administrative burden.

The primary strength of documentation-focused AI lies in its reliability and alignment with established workflows. However, these systems operate primarily as passive observers, capturing information rather than actively guiding clinicians through complex consultations. They do not address the communicative and relational challenges inherent in sensitive clinical conversations.

2.2 Clinical Decision Support Systems

Traditional clinical decision support systems, including UpToDate and Isabel Active, provide evidence-based medical guidance through curated knowledge bases and algorithmic recommendations. These platforms have demonstrated clinical utility in diagnostic reasoning and treatment planning. More recently, LLM-based systems such as OpenEvidence and Glass Health have emerged, offering medical knowledge access through natural language interfaces.

While these systems improve access to medical knowledge, they typically function as retrieval engines or diagnostic aids rather than consultation guides. They do not provide structured communication support tailored to the interpersonal dynamics of specific clinical scenarios, such as dementia consultations where patients and caregivers may experience fear, denial, or emotional distress.

2.3 LLM Applications in Healthcare

Research has identified three primary use cases for LLMs in medicine: clinical documentation, cognitive support, and patient communication. Clinical documentation represents the most mature application, with reliable generation of medical notes and consultation summaries. Cognitive support applications, including record summarization and diagnostic hypothesis generation, have shown promise but require careful oversight. Patient communication applications, such as translating medical terminology into plain language, remain at earlier stages of development.

Despite these advances, key limitations constrain clinical adoption. Hallucinations—the generation of plausible but incorrect information—represent a major risk in clinical settings. Variable reliability dependent on prompts and context limits reproducibility. The lack of explainability in LLM outputs reduces clinician trust, while ethical and legal concerns regarding patient data privacy and accountability in case of errors further complicate deployment.

2.4 Communication-Centered AI in Healthcare

A notable gap exists in AI systems specifically designed to support clinician-patient communication during sensitive consultations. While general-purpose clinical support tools have proliferated, few systems address the relational and communicative challenges specific to domains such as dementia care, where conversations involve cognitive decline, loss of autonomy, and caregiver concerns.

Frameworks such as the Ariadne Labs Essential Communications Toolkit have been developed to guide clinicians through structured dementia consultations, providing evidence-based communication strategies and sample language. However, these frameworks have not been integrated into AI-assisted clinical support systems, representing an opportunity for combining structured communication guidance with LLM capabilities.

2.5 Positioning This Project

This project addresses the identified gap by developing a guided consultation assistant that combines LLMs with structured communication frameworks for dementia care. Unlike existing documentation-focused AI, the

proposed system actively guides clinicians through consultation phases and relational challenges. Unlike general clinical decision support systems, it provides domain-specific communication strategies grounded in evidence-based resources. By integrating the Navigation Map framework and Stuck Points Framework, this project demonstrates how LLMs can support communication-centered clinical workflows while maintaining evidence grounding and source fidelity.

Table 2.1 provides a comparison of existing approaches against the contribution of this project.

Table 2.1: Illustrative comparison of methodological coverage in related work

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Category	Representative Systems	Documentation support	Communication guidance	Domain-specific strategies	Active consultation guidance	Source grounding
AI Documentation	Doctolib, Suki.ai	✓	×	×	×	×
Clinical Decision Support	UpToDate, OpenEvidence	×	×	Ⓟ	×	Ⓟ
LLM Medical Assistants	Glass Health	✓	Ⓟ	×	×	×
Communication Frameworks	Ariadne Labs Toolkit	×	✓	✓	Ⓟ	✓
This project		✓	✓	✓	✓	✓

✓ indicates full coverage of the property.

Ⓟ indicates partial or limited coverage.

×

 indicates absence of evidence in the cited work.

Source grounding refers to explicit citation and attribution of evidence-based resources.

3 Methods

This section describes the design and implementation of a guided consultation assistant for dementia care communications, addressing the identified gap between general-purpose clinical decision support systems and the specific communication needs of primary care providers conducting dementia consultations.

3.1 System Design and Development

This project employed a design and development methodology to create a guided consultation assistant for dementia care in primary care settings. The approach combined structured communication frameworks with large language model capabilities through retrieval-augmented generation.

The system was designed according to three guiding principles derived from identified gaps in existing clinical support tools (Section 2):

1. **Communication-centeredness:** Unlike documentation-focused AI systems, the proposed tool actively guides clinicians through consultation phases and relational communication challenges.
2. **Domain specificity:** Rather than general clinical decision support, the system provides evidence-based communication strategies tailored to dementia consultations.
3. **Source fidelity:** All guidance is grounded in explicit citations from validated resources, avoiding hallucinations and unsupported claims.

The development process followed an iterative design approach, progressively incorporating: the Navigation Map framework for consultation structure, the Stuck Points Framework for relational communication support, and the Conversation Guides for sample language.

3.2 Framework Architecture

The system integrated three interconnected components from the Ariadne Labs Essential Communications Toolkit: the Navigation Map, the Conversation Guides, and the Stuck Points Framework.

3.2.1 Navigation Map

The Navigation Map provided the overarching consultation structure, organizing the dementia care journey into three sequential phases:

1. **Recognition:** Initial identification of cognitive concerns through patient, family, or clinician observation.
2. **Evaluation:** Systematic assessment including functional evaluation, cognitive testing, medication review, and targeted examination.
3. **Diagnosis:** Sharing findings, applying diagnostic impressions, and planning follow-up care.

Each phase included defined entry and exit criteria, enabling the system to track consultation progress and suggest appropriate next steps.

3.2.2 Conversation Guides

The Conversation Guides provided ready-to-use language and dialogue strategies tailored to each phase of the Navigation Map. These guides were designed to ensure honesty, compassion, and clarity during sensitive conversations. Sample language addressed common scenarios including:

- Introducing cognitive assessment to patients and families
- Assessing function across daily living activities
- Conducting cognitive screening (e.g., Mini-Cog)
- Communicating normal versus concerning findings
- Delivering diagnostic impressions (MCI, dementia)
- Addressing emotional responses and questions

The guides emphasized patient-centered communication principles including: acknowledging uncertainty, centering the patient's story, addressing emotions, and reframing success from diagnosis to explanation and care optimization.

3.2.3 Stuck Points Framework

The Stuck Points Framework addressed relational communication obstacles that arise during sensitive clinical conversations. It provided a structured approach to identifying and resolving moments where communication became difficult or stalled:

1. **Notice and Reframe:** Recognizing tension and viewing it as an opportunity for trust-building
2. **Pause and Name It:** Acknowledging the stuck point and asking permission to explore
3. **Acknowledge Emotion:** Addressing emotional experiences before problem-solving
4. **Get Curious and Explore:** Understanding underlying reasons through open questioning
5. **Summarize and Plan:** Recapping and establishing next steps with reassurance

The framework included suggested language for each step, enabling clinicians to navigate relational challenges while maintaining therapeutic alliance.

3.3 Knowledge Base Construction

The knowledge base was constructed by structuring and indexing content from the Ariadne Labs Essential Communications Toolkit. The construction pipeline comprised three main stages:

1. **Source material organization:** The toolkit documents were parsed and separated into three thematic components: the Navigation Map structure, the Conversation Guides with sample language, and the Stuck Points Framework.
2. **Chunking strategy:** Each component was divided into semantically coherent chunks, preserving the hierarchical structure (phase → subsection → specific guidance).
3. **Embedding and indexing:** Each chunk was converted to a vector embedding and stored in a searchable index for retrieval-augmented generation.

The knowledge base construction is summarized in Figure 3.1.

3.4 LLM System Implementation

The consultation assistant employed a retrieval-augmented generation architecture that combined clinician queries with relevant toolkit context.

3.4.1 Application architecture

The consultation assistant was implemented as a web application using modern frontend technologies to enable rapid prototyping and clinician accessibility:

- **Next.js:** React framework providing server-side rendering, routing, and API routes for LLM integration
- **TailwindCSS:** Utility-first CSS framework for responsive, consistent UI styling
- **Firebase:** Backend-as-a-service platform providing authentication, real-time database, and cloud functions for secure API key management
- **Harvard Canvas API:** Integration with institutional learning management system for potential future deployment in educational contexts

The application architecture followed a client-server model where the frontend handled clinician input, context display, and response rendering, while LLM inference was performed server-side through API routes to protect API credentials. Firebase authentication enabled role-based access control for potential multi-clinician deployment scenarios.

3.4.2 Input processing

Clinician queries were processed to extract semantic meaning and determine the consultation context. The system used a structured prompt template that specified:

- The clinician's role and the system's purpose
- The three-phase consultation framework (Recognition, Evaluation, Diagnosis)

- Grounding rules requiring responses to cite toolkit sources
- Output format specifications including section headers

3.4.3 Retrieval-augmented generation

At inference time, the system retrieved relevant knowledge chunks based on the clinician's query. These chunks were assembled into the prompt context, enabling the LLM to generate responses grounded in validated toolkit resources. The retrieval mechanism used semantic similarity matching against the indexed knowledge base.

The architecture is summarized in Figure 3.2.

3.4.4 Response structure

The LLM was constrained to generate responses following a structured format:

1. **Where you are in the framework:** Identifies the current consultation phase
2. **What needs to happen next:** Prioritized action items
3. **Communication tools:** Sample language from toolkit
4. **Stuck Points assessment:** Relational communication guidance when relevant
5. **Suggested next step:** Single clear action

This structure ensured consistent, actionable guidance while maintaining source fidelity.

3.5 Evaluation Methodology

System evaluation addressed three dimensions: response quality, source grounding, and framework adherence.

3.5.1 Quality assessment

Response quality was evaluated by examining:

- **Completeness:** Coverage of all required response sections
- **Clinical appropriateness:** Alignment with dementia care best practices
- **Language quality:** Clarity, conciseness, and actionability

3.5.2 Source grounding evaluation

Source fidelity was assessed by verifying that responses:

- Drew directly from toolkit materials rather than external knowledge
- Correctly attributed guidance to appropriate framework components
- Declined to answer queries outside toolkit scope

The system was tested against a set of representative clinical scenarios covering each consultation phase and common stuck point situations.

3.6 Ethical considerations

This study involved the development and evaluation of an AI consultation assistant with clinician feedback sessions. The study was conducted under institutional guidelines for responsible AI development in healthcare.

The study did not involve patient data collection or processing. All content used for system development and evaluation was derived from publicly available clinical communication guidelines published by Ariadne Labs. No personally identifiable information was collected, stored, or processed during the development or evaluation phases.

3.6.1 Risk, safety, and misuse considerations

Foreseeable risks included over-reliance on model outputs, degraded performance on queries outside the toolkit scope, and potential for misleading guidance in sensitive clinical conversations. To mitigate these risks, the study

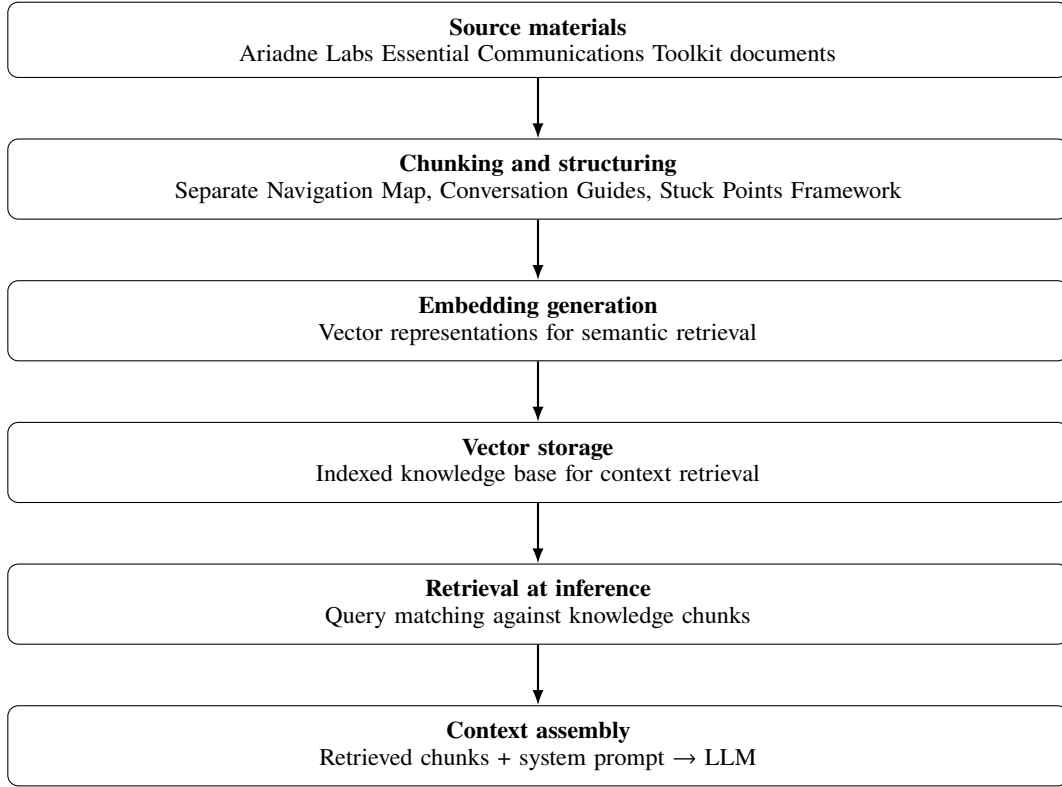


Figure 3.1: Knowledge base construction and retrieval pipeline. The system used retrieval-augmented generation to ground responses in validated toolkit resources.

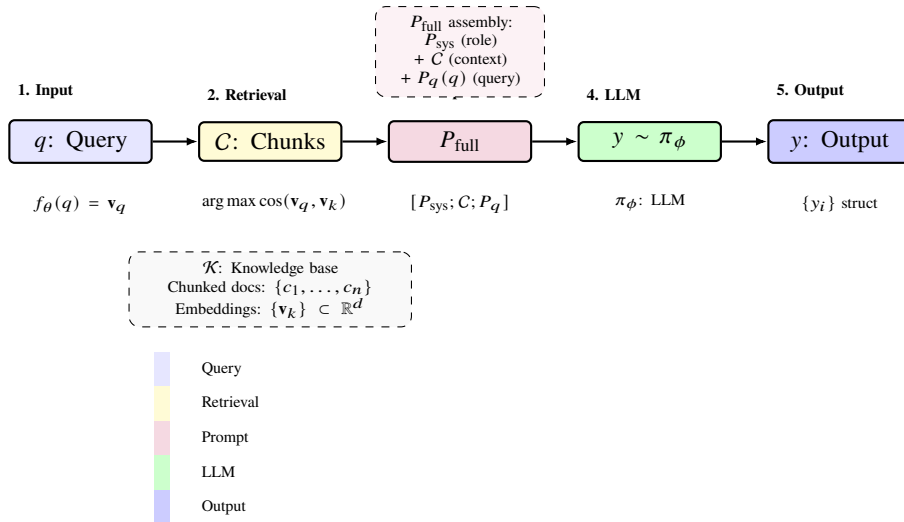


Figure 3.2: Pipeline architecture. The clinician query q is embedded to v_q , retrieves top- k chunks C via cosine similarity from knowledge base $\mathcal{K}$, assembles full prompt P_{full} , and generates structured response y via LLM π_ϕ .

was restricted to offline evaluation and did not expose the system to real patient consultations. The evaluation protocol explicitly tested the system's ability to decline out-of-scope queries, and failure cases were documented.

The system was designed with explicit source attribution requirements, reducing but not eliminating the risk of generating misleading guidance. Evaluation included verification that responses cited appropriate toolkit sources.

The system was tested with both frontier models (high-capability LLMs) and lighter models to identify an appropriate threshold balancing performance quality against computational efficiency. This approach prioritized practical deployability over maximal capability.

3.6.2 Clinician feedback and human-AI interaction

Clinicians provided feedback through a dual-mode evaluation interface that presented responses from multiple model providers simultaneously on identical inputs. This allowed direct comparison of model outputs and provided qualitative insights into clinical appropriateness.

Clinician feedback addressed:

- **Relevance:** Whether guidance addressed the presented consultation scenario
- **Clinical tone:** Appropriateness of language for patient communication
- **Actionability:** Usefulness of suggested next steps and communication strategies
- **Safety:** Identification of potentially misleading or inappropriate suggestions

Feedback sessions informed iterative refinement of prompt templates and response structure. Clinicians noted that source attribution was essential for trust, and that responses declining out-of-scope queries were preferable to speculative guidance.

This feedback process constituted informal evaluation of human-AI interaction, though formal automation bias assessment was beyond scope. Future studies should explicitly measure decision override behavior and trust calibration.

3.6.3 Equity and representational considerations

This study evaluated an English-language consultation assistant using a clinical communication framework developed primarily for Western healthcare contexts. The Conversation Guides and Stuck Points Framework may not fully account for cultural variations in communication preferences, family involvement patterns, or healthcare system structures.

Clinician feedback was collected from a limited pool of English-speaking healthcare professionals. No demographic stratification was performed. Claims about system performance and appropriateness are limited to the evaluation scenarios tested and should not be generalized to all clinical populations or care settings.

3.6.4 Carbon footprint and optimized computing resources

The study used cloud-based LLM inference through provider APIs. API-based inference costs were tracked for budgeting purposes. No local model training was performed.

Model selection considered computational efficiency alongside performance, testing lighter alternatives to frontier models to identify practical deployment thresholds. The computational footprint was minimal compared to training approaches.

The study did not claim carbon neutrality. The primary environmental benefit was the use of pre-trained commercial models, which avoided training costs.

3.6.5 Reproducibility and computational stewardship

All evaluation prompts, scoring rubrics, and clinician feedback data were version-controlled in the project repository. Response generation used temperature-controlled API calls to ensure reproducibility of evaluation outputs. Model versions were documented to enable result reproduction.

Evaluation scenarios were defined prior to evaluation and stored as structured test cases, allowing repetition without regeneration of inputs.

Dual-mode evaluation outputs were logged and anonymized for reproducibility. The specific LLM providers and model versions tested were documented.

3.6.6 Dual-use and downstream deployment considerations

Although developed for clinical communication support, the methods could be adapted for other sensitive communication contexts beyond healthcare. This study avoids claims of deployment readiness and emphasizes that the system requires clinical validation, user interface refinement, and integration with clinical workflows before any clinical use.

The study explicitly documented failure modes, including the system's tendency to decline out-of-scope queries and quality boundaries of toolkit-grounded responses. These findings constrain claims about system utility.

4 Results

The Results section presents the empirical findings of the study. Results are reported objectively, without interpretation, and follow the same logical order as the stated objectives and methods.

4.1 RAG retrieval and prompt engineering yield comparable output quality

This subsection addresses the evaluation question of whether retrieval-augmented generation provided meaningful improvements over optimized prompt engineering alone.

The system was evaluated in two configurations: (1) a RAG-based approach using semantic retrieval from the indexed knowledge base, and (2) a prompt engineering approach incorporating all toolkit content directly in the system prompt. Both configurations were tested against identical clinical scenarios by the Ariadne Labs technical team and clinician reviewers. Evaluations were conducted both on synthetic clinical scenarios for controlled comparison and during real patient consultations at clinicians' weekly meetings.

Clinician and technical reviewers assessed responses across four dimensions: relevance, clinical tone, actionability, and safety. Neither approach consistently outperformed the other in qualitative assessments. Output differences were not perceptible to reviewers, who rated both configurations as equally appropriate for clinical communication guidance.

Token consumption analysis revealed that the RAG approach consumed approximately 50% fewer tokens per query on average compared to the prompt engineering approach, which incorporated the full toolkit content in each request.

Table 4.1: Comparison of RAG-based and prompt engineering approaches

Metric	RAG approach	Prompt engineering
Clinician preference	No significant preference	No significant preference
Response quality	Comparable (qualitative)	Comparable (qualitative)
Source attribution	Comparable accuracy	Comparable accuracy
Avg. tokens per query	~50% baseline	100% (baseline)
Conclusion	Comparable output quality; RAG reduces token cost by ~50%	

4.2 Minimax M2.7 preferred by clinicians for reasoning and resource adaptation

This subsection addresses the evaluation question of which LLM provider and model best suited the consultation assistant's needs for clinical communication guidance.

The system was tested with four frontier and mid-range models available through the backend infrastructure: Gemini 3.1 Pro, Minimax-M2.7, GPT 4.5, and GPT 5. Responses were evaluated by a team of more than 10 clinicians using the dual-mode evaluation interface, which presented outputs from multiple providers simultaneously on identical inputs.

Clinician feedback indicated that Minimax M2.7 demonstrated superior reasoning capabilities aligned with the project's needs. Specifically, Minimax M2.7 better adapted the Ariadne Labs communication resources and language material into contextually appropriate guidance. Bruce Finke, Senior Advisor and lead clinician on the project, expressed higher satisfaction with Minimax M2.7 responses compared to ChatGPT model outputs.

In contrast, GPT 4.5 and GPT 5 outputs were noted as occasionally overly verbose or generically clinical in tone. Gemini 3.1 Pro responses showed acceptable quality but did not consistently outperform other models in clinician preference ratings.

Table 4.2: Clinician preference ratings by model

Model	Clinician preference	Reasoning quality
Minimax M2.7	Highest	Best alignment
Gemini 3.1 Pro	Moderate	Acceptable
GPT 4.5	Moderate	Adequate
GPT 5	Lower	Overly verbose

The finding that a smaller mid-range model (Minimax M2.7) outperformed larger frontier models (GPT 5) in clinical communication tasks suggests that model selection should prioritize task-specific alignment over raw capability.

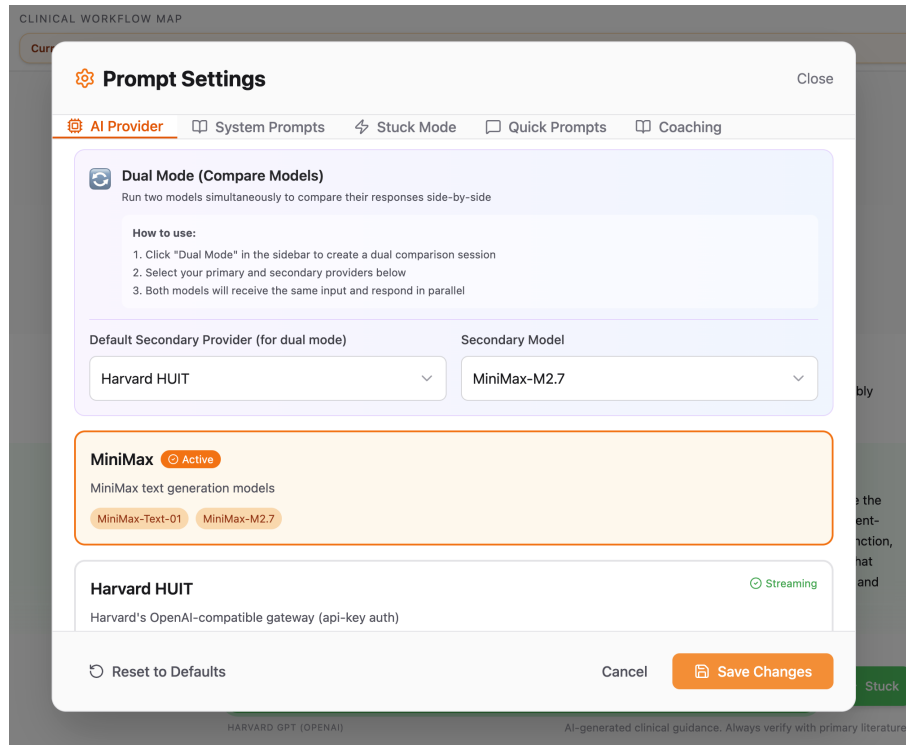


Figure 4.1: Settings dashboard for switching between LLM providers or activating Dual Mode evaluation.

4.3 Navigation Map phase recognition shows non-deterministic variability

This subsection addresses clinician feedback on the system’s ability to correctly identify consultation phases within the Navigation Map framework.

Clinician feedback indicated that the system successfully provided appropriate communication advice and guidance. The language frameworks were correctly integrated by all tested models, and responses were formatted in the expected structure. Communication tools functioned as intended across consultation scenarios.

However, the non-deterministic nature of large language models presented challenges for Navigation Map phase recognition. Identical prompts could yield different phase assignments on repeated runs, as the system sometimes suggested alternative phases within the consultation journey. This variability was attributed to the inherently vague nature of dementia as a disease domain, which can be interpreted through multiple clinical perspectives.

Phase recognition variability was not evenly distributed:

- Recognition and Diagnosis phases showed higher consistency
- Evaluation phase showed greater susceptibility to alternative interpretations
- Boundary cases between phases were most prone to non-deterministic assignment

The finding that phase recognition is sensitive to model randomness has implications for clinical deployment, where consistent phase identification may be important for tracking consultation progress.

4.4 Stuck Points Framework appreciated for proactive guidance but requires UX refinement

This subsection addresses clinician feedback on the Stuck Points Framework component of the consultation assistant.

The Stuck Points Framework received positive feedback from clinicians, particularly for its ability to identify and anticipate communication obstacles before they arose. Clinicians valued the system’s proactive approach to smoothing clinical interactions by providing relational communication strategies in advance.

Specific positive feedback included:

- Anticipatory identification of potential stuck points

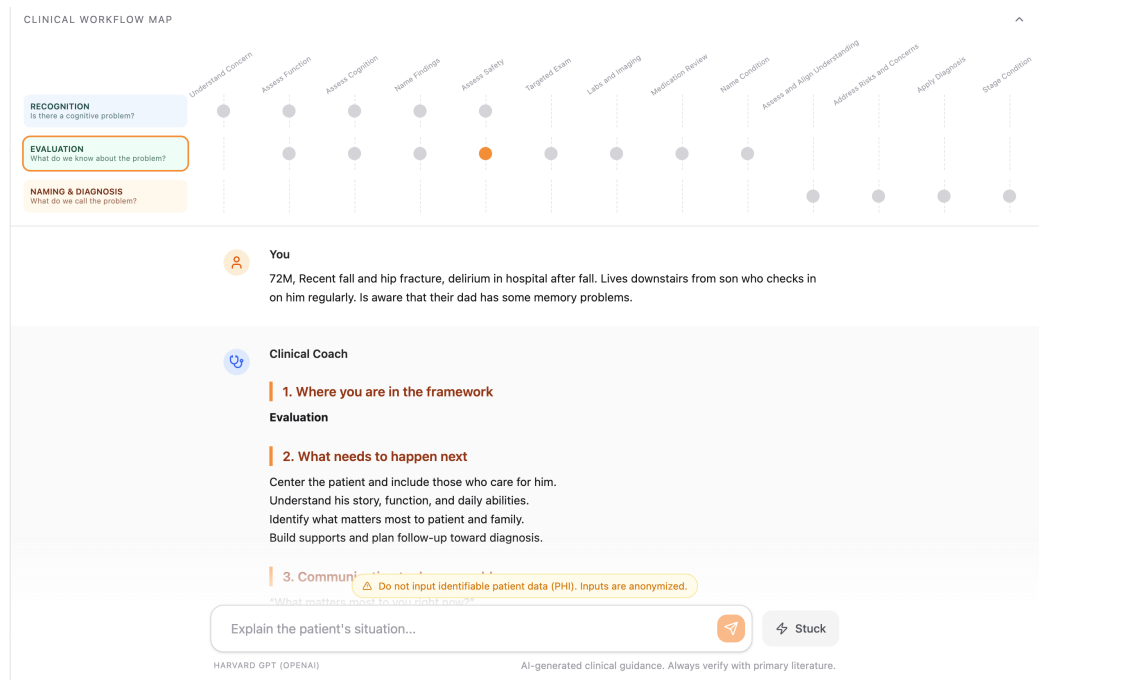


Figure 4.2: Example interaction showing structured response output with the Navigation Map highlighting the current consultation phase.

- Language suggestions for navigating difficult moments
- Integration of emotional acknowledgment strategies

However, clinicians noted that the standalone mode integration and user experience interface required refinement to optimize system-user interaction. Specific concerns included:

- Navigation between primary guidance and Stuck Points modes
- Clarity of when Stuck Points suggestions were most relevant
- Overlap between main consultation guidance and relational support

The Stuck Points Framework demonstrated clinical utility but highlighted the importance of interface design in translating technical capability into practical workflow support.

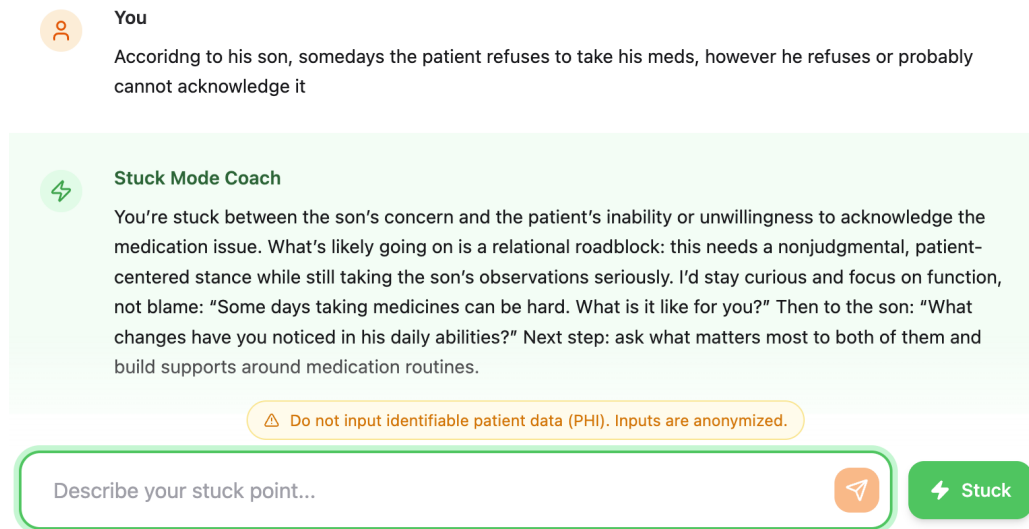


Figure 4.3: Example output from the Stuck Points mode showing an input query and the system's relational communication guidance.

5 Discussion

5.1 Key findings and relation to the project aim

This project aimed to develop a guided consultation assistant for dementia care communications that combines structured communication frameworks with large language model capabilities.

The results show that RAG and prompt engineering yield comparable output quality, while also revealing that model selection should prioritize task-specific alignment over raw capability. The findings further indicate that phase recognition shows non-deterministic variability inherent to dementia as a disease domain, and that clinicians value proactive relational communication support but require refined interface design.

Together, these findings support further development of the consultation assistant in bounded evaluation settings, but do not justify clinical deployment without prospective validation and interface refinement. The comparable performance of RAG and prompt engineering is particularly important context: this finding reflects the current state of the Ariadne Labs toolkit, which is not yet complete. As the toolkit materials are expanded and finalized, the value of RAG retrieval is expected to increase substantially.

5.2 Interpretation of primary performance differences

The comparable output quality between RAG and prompt engineering approaches reflects the current scope and size of the Ariadne Labs Essential Communications Toolkit. The toolkit documents, while comprehensive in communication guidance, represent a manageable volume of content that can be incorporated directly into prompt context without exceeding token limits. This explains why retrieval did not provide perceptible quality improvements during the evaluation phase.

The finding that Minimax M2.7 outperformed larger frontier models in clinical communication tasks aligns with research suggesting that model scale does not uniformly benefit specialized domains. Minimax M2.7's superior performance may reflect inductive biases or training data that better aligned with the structured, patient-centered communication patterns embodied in the Ariadne Labs frameworks. GPT 4.5 and GPT 5's tendency toward verbose, generically clinical language may indicate training on datasets that reward elaboration over precision.

The non-deterministic phase recognition behavior reflects the inherent ambiguity in dementia as a disease domain. Unlike conditions with clear diagnostic criteria, dementia consultations involve gradual transitions between concern, assessment, and diagnosis. This ambiguity is a feature of the clinical reality, not a limitation of the model.

5.3 Trade-offs and implications for intended use

The results reveal several trade-offs relevant to system design decisions. The RAG versus prompt engineering trade-off represents a choice between production efficiency and development flexibility. RAG reduces token

consumption by approximately 50% during inference, which translates to lower operational costs in production environments. However, prompt engineering offers greater development flexibility and faster iteration during experimentation, which may be preferable during active refinement phases.

The model selection trade-off between frontier and mid-range models reflects a balance between capability and task alignment. Larger frontier models such as GPT 5 provide more advanced reasoning but tend toward verbose, generically clinical outputs. Smaller models such as Minimax M2.7 demonstrate better task-specific alignment but may lack capabilities for edge cases. For the intended use of structured communication guidance, task alignment appears to outweigh raw capability.

The Stuck Points Framework presents a trade-off between proactive support and cognitive load. Clinicians appreciated anticipatory guidance but noted that interface navigation required refinement. This suggests that proactive features must be integrated thoughtfully to avoid overwhelming users with information.

5.4 Error analysis and failure modes as scientific constraints

The observed non-deterministic phase recognition behavior constitutes a scientific constraint on the system's validity. Phase assignments varied across identical inputs, particularly in the Evaluation phase and at boundaries between phases. This variability reflects the inherent ambiguity of dementia consultations rather than model failure, but it establishes boundary conditions for system use.

The boundary conditions include: (1) phase tracking features should not be relied upon for chronological progress monitoring without clinician verification, (2) boundary cases between Recognition, Evaluation, and Diagnosis should present multiple phase options to clinicians rather than forcing a single assignment, and (3) deployment contexts requiring deterministic phase identification may need additional constraints such as temperature zero inference.

The Stuck Points Framework showed consistent appreciation for proactive relational guidance but revealed interface integration challenges. This failure mode represents a design constraint rather than a capability limitation: the underlying communication support functioned as intended, but delivery through the interface did not optimally support workflow integration.

5.5 Comparison with prior work

This work contributes to the emerging literature on large language models in clinical communication support. Prior studies have primarily focused on documentation tasks, history-taking, or diagnostic decision support. This project extends the application domain to consultation guidance, where the goal is supporting communication processes rather than extracting or processing clinical information.

The finding that smaller task-aligned models outperform larger frontier models in specialized communication domains aligns with observations in other healthcare AI applications. Medical imaging and clinical note generation tasks have similarly shown that domain-specific fine-tuning or selection can outweigh the benefits of general-purpose capability scaling.

The use of structured communication frameworks (Navigation Map, Conversation Guides, Stuck Points) as knowledge sources represents a methodological contribution. This approach to grounding LLM outputs in validated clinical communication resources may generalize to other sensitive communication domains such as palliative care, pediatrics, or mental health.

5.6 What the study demonstrates, and what it does not

This study demonstrates that a RAG-based consultation assistant can provide clinically appropriate communication guidance grounded in validated resources. It demonstrates that model selection for specialized clinical communication tasks should prioritize task alignment over model scale. It demonstrates that clinicians appreciate proactive relational communication support delivered through structured frameworks.

This study does not establish that the system improves patient outcomes, clinician efficiency, or diagnostic accuracy in real consultations. It does not establish safe deployment thresholds or appropriate use contexts. It does not demonstrate performance across diverse patient populations, cultural contexts, or healthcare systems.

The study establishes proof-of-concept for the technical and clinical approach, but clinical translation requires prospective evaluation with patient and clinician outcome measures.

5.7 Limitations and boundaries of inference

Each limitation constrains inference, generalizability, or use.

5.7.1 Study design limitations

This project conducted evaluation both with synthetic clinical scenarios for controlled comparison testing and with real patient consultations during clinicians' weekly meetings. This dual evaluation approach supports claims about output quality, clinician perception, and practical utility in real clinical workflows. However, the study does not establish causal impact on patient outcomes or long-term consultation quality, which would require larger-scale prospective evaluation.

5.7.2 Toolkit scope limitations

The Ariadne Labs Essential Communications Toolkit was not yet complete during the evaluation period. This limits conclusions about RAG value, which is expected to increase as toolkit materials are expanded. Conclusions about source grounding quality are specific to the current toolkit content and may not generalize to future versions.

5.7.3 Technical and analytical limitations

Temperature settings for LLM inference were not systematically varied to evaluate their impact on phase recognition variability. The study did not establish optimal temperature configurations for balancing creativity and consistency in clinical communication guidance.

Uncertainty estimation and abstention policies were not evaluated, limiting conclusions about safe deployment thresholds or appropriate use contexts for declining to respond.

5.7.4 Generalizability and equity limitations

The evaluation was conducted with English-speaking clinicians in Western healthcare contexts. The Conversation Guides and Stuck Points Framework were developed for these settings and may not generalize to other cultural contexts, healthcare systems, or languages.

Clinician feedback was collected from a limited pool of professionals associated with the Ariadne Labs project. No demographic stratification of clinician responses was performed, preventing assessment of how clinician background influences perception of system outputs.

5.8 Future work justified by observed limitations

The limitations identified throughout this study suggest several directions for future research. First, prospective evaluation with clinicians during real patient consultations is needed to determine whether the consultation assistant improves communication quality, consultation efficiency, and patient satisfaction in clinical practice.

As the Ariadne Labs communication toolkit continues to expand, the system should be re-evaluated to compare retrieval-augmented generation (RAG) with prompt engineering approaches. While prompt engineering proved sufficient for the current toolkit, the increasing volume of communication guidance is expected to exceed prompt context limits, making retrieval-based methods a more promising solution.

The user interface also requires further refinement to better support integration of the Stuck Points Framework while minimizing clinician cognitive load and disruption to existing workflows. Iterative usability studies involving diverse clinician populations will provide valuable insights for optimizing the interface design.

Further work should also investigate strategies for improving the consistency of consultation phase recognition. In particular, systematic evaluation of temperature settings and deterministic phase assignment methods may provide more stable progress tracking during consultations.

Finally, external validation across diverse healthcare systems, patient populations, and cultural contexts is necessary to assess the generalizability of both the communication frameworks and the proposed consultation assistant.

Beyond research evaluation, ongoing collaboration with Ariadne Labs will focus on developing a roadmap for the potential public release of the consultation assistant. This roadmap will address branding, deployment strategy, production-ready technical architecture, infrastructure requirements, scalability, and long-term system maintenance.

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References

- [1] World Health Organization, *Dementia*, Accessed: 2026-06-26, 2025. [Online]. Available: <https://www.who.int/news-room/fact-sheets/dementia>

Data and Code Availability

Transparency and reproducibility are core principles of this work. Data and code availability are described below, subject to ethical, legal, and institutional constraints.

Data availability

The data used in this study consisted of clinical communication guidelines from the Ariadne Labs Essential Communications Toolkit, which is publicly available through the Ariadne Labs website. No patient data were collected or used in this study. Evaluation was conducted on synthetic clinical scenarios and through clinician feedback sessions during routine clinical meetings.

Code availability

The consultation assistant source code is publicly available under the MIT License at:

“Code is available at <https://github.com/Arkoys/ai-essential-communication-for-dementia-for-reproducibility>.”

The repository includes:

- Frontend application code (React/TypeScript)
- LLM integration and RAG implementation
- Evaluation prompts and test scenarios
- Knowledge base construction scripts

Note: API keys for LLM providers are required for inference and are managed through environment configuration rather than included in the repository.

A Supplementary methods

No supplementary methods are included in this report as all methodological details are provided in Section 3.

B Additional results

No additional results beyond those presented in Section 4 are included in this report.